

Comparing Exploration Algorithms in the 3D Maze

A Course-Project Reproduction under the Paper’s Appendix C.2 Settings

Challenging Problems in Reinforcement Learning (SS 2026)

Topic: Exploration vs. Exploitation

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results updated 2026-06-03: LPM corrected to the paper’s paper-faithful log-space reward (Eq. 3); Appendix-C.2 config ($\lambda_i=1$, entropy 0.03, raw reward); 6 policies $\times$ 3 variants $\times$ 64 seeds

Abstract

We compare five exploration mechanisms—Learning Progress Monitoring (LPM), Random Network Distillation (RND), the Intrinsic Curiosity Module (ICM), a raw next-state prediction-error bonus (MSE), and a no-intrinsic A2C baseline (NONE)—plus a *uniform-random control* (no learning at all) in the Miniworld 3D maze of Hou, An & Du (2026), *Beyond Noisy-TVs: Noise-Robust Exploration via Learning Progress Monitoring*. The maze emits *no extrinsic reward*, so a method’s quality is its *spatial coverage*, and the noisy-TV failure mode is geometric: a random-RGB “noise wall” at $z \approx 8$ separates the two lower spawn rooms from the far room. **This version corrects LPM to the paper’s faithful *log-space* reward $r^i = g_\phi - \log \text{MSE}$ (Eq. 3), replacing the released notebooks’ raw-error clipped form; doing so also required lowering the error-model learning rate (the notebook value silently broke the log-space objective).** We run the paper’s exact Appendix C.2 configuration ($\lambda_i=1$, entropy 0.03, raw intrinsic reward) at the paper’s budget but with 64 seeds (5×10^4 steps each). Three findings stand out. **(1)** The paper’s central claim—*noise robustness*—reproduces cleanly with the faithful reward: under action-triggered noise, LPM selects the noisy-TV action only 3.0% of the time (median 0.9%; 1/64 seeds) while the raw-pixel method (MSE) is trapped on it 83% of the time and its coverage collapses. **(2)** On *coverage*, no learned method beats the pure uniform-random control: uniform covers the most in *all three* variants. In this small maze coverage rewards undirected randomness, so the paper’s “LPM achieves the highest coverage” ranking does *not* reproduce, while its noise-robustness claim—decoupled from coverage—does. **(3)** The faithful, *unclipped* $\lambda_i=1$ reward is noise-robust but *unstable*: 45% of LPM’s **action noise** seeds collapse to degenerate near-stationary policies, pushing LPM’s coverage below ICM in both mean and median—a reversal from the earlier clipped run. The collapse is policy instability, *not* TV-fixation (the collapsed seeds still avoid the TV), and the notebook’s clip was masking it. A 64-seed-averaged, log-scaled occupancy-density figure renders the spatial dynamics.

1 Motivation and research questions

The course topic is the exploration–exploitation trade-off. Deep RL drives directed exploration with intrinsic-motivation bonuses (prediction error, novelty, learning progress). The *noisy-TV* problem is the canonical failure mode of prediction-error exploration: an unlearnable source of randomness produces perpetually high prediction error, so a curiosity-driven agent fixates on it. LPM rewards the *improvement* of its dynamics model rather than the error level, and is therefore noise-robust by construction. The 3D maze makes this failure spatial and directly visualisable: the noise wall is a physical place the agent can get stuck at.

The maze geometry is a hand-designed four-room layout in an 18×12 world (continuous units): two spawn rooms below the wall (room1 $x \in [0, 4]$, room2 $x \in [14, 18]$, both $z \in [0, 8]$), the thin grass-textured *noise wall* (room3, $z \in [8, 8.1]$), and the large far room (room4, $z \in [8.1, 12]$). The agent spawns in room1 and must traverse the wall to explore room4.

RQ1 (coverage & noise robustness, reproduction).

Under the noisy variants (`noisy.tv`: random-RGB wall + 25% sticky actions; `action_noise`: an action that shows a random CIFAR-10 image) vs. the clean control (`nonoise`), which methods keep expanding coverage and which stall—and crucially, do any of them beat a *uniform-random* policy that does no learning at all?

RQ2 (spatial dynamics, novelty).

How does the spatial distribution of exploration evolve over training? We answer this with a coverage-heatmap-evolution figure (rows: method; columns: training-time window) *averaged over all 64 seeds*.

2 Methods

All learned methods use the same A2C base, configured to the paper’s Appendix C.2 (MiniWorld): RMSprop, policy LR 0.01, $\gamma = 0.99$, GAE $\lambda = 0.95$, value-loss coef. 0.5, gradient clipping 0.5, a single-epoch update every 64 steps, dynamics-model LR 10^{-3} , error-buffer size 100, entropy coefficient 0.03, and intrinsic weight $\lambda_i = 1$, over a Nature-style CNN on 160×120 RGB frames. Because the maze extrinsic reward is identically zero, the per-step training reward is the *raw* intrinsic term,

$$r_t = r_t^{\text{ext}} + \lambda_i r_t^i = \lambda_i r_t^i, \quad \lambda_i = 1, \quad (1)$$

with no running-std normalisation. (The upstream notebooks instead used $\lambda_i = 0.1$, entropy 0.05, and a normalised reward—settings that *disagree* with the paper’s own Appendix C.2 and make the intrinsic signal $10\times$ too weak; we follow C.2. The deviation is logged in `UPSTREAM.md`.) For `NONE`, $r_t \equiv 0$, so the A2C policy is still *trained* (by the entropy bonus and value bootstrap) but receives no reward signal; the *uniform* control is not trained at all—it samples actions $a \sim \text{Uniform}\{0, \dots, 4\}$ every step.

LPM corrected to the paper’s log-space reward. The released maze notebooks (and our earlier runs) computed the LPM bonus in *raw*-MSE space with a clip, $r^i = \min(0.5, \eta e^{g_\phi} - \text{MSE})$ with $\eta=1$, which does *not* match the paper’s Eq. (1)–(3). We corrected LPM to the faithful log-space form: with the per-step error defined as $\varepsilon = \log \text{MSE}$ (Eq. 1) and an error model g_ϕ trained to predict the *previous* dynamics model’s log-error $\mathbb{E}_{\mathcal{D}}[\varepsilon^{(\tau-1)}]$ (Eq. 2), the intrinsic reward is their difference (Eq. 3),

$$r_t^i = g_\phi(o_t, a_t) - \varepsilon_t^{(\tau)} = \mathbb{E}_{\mathcal{D}}[\log \text{MSE}_{\text{prev}}] - \log \text{MSE}_{\text{cur}}, \quad (2)$$

with no η and no clip, gated to 0 until the error buffer fills ($|\mathcal{D}|=100$, Alg. 1), and g_ϕ updated together with the dynamics model every step. This also required lowering the error-model learning rate to 10^{-3} : under the log-space objective the notebook’s 10^{-2} drives g_ϕ into its $[-10, 10]$ clamp (zero gradient—a dead error model that pins the reward at a constant ≈ -6); at 10^{-3} the error model fits the log-error targets and the bonus becomes the intended learning-progress signal, fluctuating around 0 and decaying as the model saturates (cf. the paper’s Fig. 2). The pre-fix raw form is preserved behind `reward_space="raw"`; the fix and its root-cause are logged in `UPSTREAM.md`.

Class	Method	Mechanism (intrinsic reward r^i)
Learning progress	LPM	$g_\phi - \varepsilon = \mathbb{E}_{\mathcal{D}}[\log \text{MSE}_{\text{prev}}] - \log \text{MSE}_{\text{cur}}$ (Eq. 2): expected-minus-actual <i>log</i> -error (model improvement)
Prediction error	RND	error of a predictor vs. a fixed random target network on the next frame
Prediction error (feat.)	ICM	forward-dynamics error in an inverse-dynamics feature space
Prediction error (pixel)	MSE	raw next-state pixel-prediction error (the pure noisy-TV victim)
None (baseline)	NONE	no intrinsic reward; A2C <i>is</i> still trained (entropy + value bootstrap)
Control	uniform	$a \sim \text{Uniform}\{0 \dots 4\}$; <i>no learning</i> (pure random walk)

Table 1: The six policies. LPM and MSE share the same pixel-decoder predictor, isolating the effect of the learning-progress signal vs. raw error. ICM and RND are canonical implementations added on the shared CNN encoder. The *uniform* control is the proper “does any of this beat chance?” baseline—distinct from NONE, whose policy is still shaped by training.

3 Experimental design

A single manipulation grid: policy $\times$ variant $\times$ seed.

Grid. Six policies (LPM, RND, ICM, MSE, NONE, uniform) $\times$ three variants (`nonoise`, `noisy_tv`, `action_noise`) $\times$ **64 seeds** = 1152 runs. Each run is a single environment trained for 5×10^4 steps—the paper’s maze budget (Fig. 3 of Hou et al. trains for 5×10^4 exploration steps). We use 64 seeds (the paper used 10) to tighten the estimates and resolve LPM’s bimodal seed behaviour.

Budget. The runs are independent, so the grid is embarrassingly parallel: on a 128-core Linux box, `run_grid.py --jobs 120 --threads-per-job 1` finished the full 1152-run grid in ≈ 6 h at $\sim$ full CPU (0 failures; peak ~ 140 GB of 1.1 TB RAM, set by the 120-way concurrency). Single-environment A2C is a batch-size-1 forward pass, so one CPU thread per job is optimal (two threads gave only $1.25\times$ per run)—the speed-up comes from running many processes. Runs use CPU (the box has no GPU); the 160×120 frames render headless via Mesa EGL (`PYGLET_HEADLESS=true`, `LP_NUM_THREADS=1`). The grid is resumable—a run’s position `.npz` marks it complete.

4 Metrics

Coverage is measured on the maze’s native 72×48 occupancy grid (cells of 0.25 world units), restricted to the *reachable* cells (the union of the four rooms; the empty corridor between the spawn rooms is excluded). Let the final value of a quantity be its mean over the last 10% of logged updates, reported as mean $\pm$ std over the 64 seeds.

- **Total coverage:** $C = (\text{visited reachable cells})/(\text{reachable cells})$, reported both as a final number and as a learning curve vs. frames.

- **TV-action share:** fraction of steps spent on the noisy-TV action (action 4) under `action_noise`—the direct measure of noisy-TV fixation. Prediction: highest for the pixel prediction-error method (MSE), lowest for LPM, exactly 1/5 for the uniform control.
- **Coverage-heatmap evolution (averaged):** for each variant, a rows(method) $\times$ columns(training-window) grid of top-down occupancy heatmaps *averaged over all 64 seeds*. The primary view is *occupancy density* = the mean per-cell visit count, drawn on a single *shared log colour scale* across all panels (a colorbar is overlaid). The raw per-cell count spans ~ 0.1 to $\sim 10^3$ —a few dwell/spin cells are visited $1000\times$ while typical cells see 1–5 visits—so a per-panel *linear* scale (our earlier figures) washed the path structure out; the shared log scale exposes both *where* each policy dwells and how much. A companion *coverage-frontier* view (fraction of seeds that ever visited each cell, 0–1) is also produced. The maze rooms and the $z = 8$ wall line are overlaid.

5 Results

We ran the full 1152-run grid at 5×10^4 steps under the C.2 configuration with the corrected log-space LPM. The single-environment A2C is high-variance (between-seed std 0.08–0.32), so we flag which conclusions are robust.

Noise robustness reproduces—cleanly, even with the faithful reward. In `action_noise`, action 4 replaces the observation with a random CIFAR-10 image. The corrected log-space LPM selects it on only $3.0\% \pm 11.3\%$ of steps (*median* 0.9%; only 1 of 64 seeds ever fixates), because its dynamics model cannot *improve* on the unlearnable image, while the raw-pixel predictor MSE is trapped on it $83\% \pm 27\%$ of the time and its coverage collapses to 0.13 (Table 2, Fig. 1). The curiosity baselines are partially fooled (RND 67%, ICM 53%). The uniform control sits at exactly 0.200 ± 0.003 (the 1/5 sanity check). This is the paper’s central claim and it reproduces.

Policy	TV-action share (<code>action_noise</code>)
LPM	0.030 ± 0.113
uniform	0.200 ± 0.003
NONE	0.205 ± 0.175
ICM	0.526 ± 0.242
RND	0.666 ± 0.276
MSE	0.830 ± 0.266

Table 2: Noisy-TV fixation under `action_noise`, mean $\pm$ std over 64 seeds. LPM essentially never stares at the noisy TV (3.0% mean, 0.9% median, 1/64 seeds fixate); MSE is trapped on it (83%); the curiosity baselines are partially fooled; the uniform control is at chance (1/5).

Coverage: no learned method beats a uniform-random policy. The headline coverage result is negative and important (Table 3, Fig. 2): the *uniform-random* control covers the most in *all three* variants—and with by far the lowest variance (std 0.08–0.13 vs 0.19–0.32 for the learned methods); in clean `nonoise` uniform (0.587) now edges even the best learned method (ICM 0.542). No learned policy, LPM included, exceeds the no-learning control anywhere. Training *reduces* coverage here: any learned policy (intrinsic-driven or the trained NONE) commits to a narrower, repetitive walk, whereas uniform thrashing maximally spreads through a small maze. Coverage in

this setting therefore rewards *randomness*, not directed exploration, and the paper’s “LPM achieves the highest coverage” ranking does *not* reproduce. Note that noise-avoidance and coverage are *decoupled*: under `action_noise` the uniform policy wastes 20% of its steps frozen on the TV yet still covers most (0.696), while LPM almost never touches the TV (3.0%) yet covers far less on average (0.299). (Uniform covers *more* under the noisy variants than under clean `nonoise` because those variants add 25% sticky actions, which create forward momentum and carry a random walker further.)

The faithful $\lambda_i=1$ reward is noise-robust but *unstable*. With the clip removed, the log-space bonus at $\lambda_i=1$ is strong, and LPM’s behaviour is sharply *bimodal*: under `action_noise`, 29 of 64 seeds (45%) collapse to a degenerate near-stationary policy (<0.05 coverage) while the rest reach 0.3–0.8. This pulls LPM’s coverage *below* ICM in both mean (0.299 vs 0.410) *and* median (0.101 vs 0.416)—a reversal from the earlier clipped-reward run, where the clip damped the reward and fewer seeds collapsed. Crucially this is *policy instability*, not *noisy-TV susceptibility*: the collapsed LPM seeds freeze in a tight loop *somewhere in the maze*, not on the TV (LPM’s TV-share stays at 3%, versus ICM’s 53%), so the collapse lowers coverage without breaking noise-robustness. The clip in the released notebooks was therefore silently *masking* a stability problem of the unclipped $\lambda_i=1$ reward; the faithful formula exposes it. This is a concrete stability-ablation hook (reward scaling λ_i , reward/return normalisation, or an advantage clip), and it reinforces that coverage—the metric LPM is built for—is the wrong yardstick in this maze.

Policy	nonoise	noisy_tv	action_noise
uniform	0.587 (0.13)	0.710 (0.08)	0.696 (0.09)
NONE	0.459 (0.24)	0.611 (0.25)	0.589 (0.25)
LPM	0.332 (0.25)	0.452 (0.27)	0.299 (0.32)
MSE	0.476 (0.23)	0.551 (0.26)	0.129 (0.12)
ICM	0.542 (0.20)	0.536 (0.29)	0.410 (0.19)
RND	0.468 (0.22)	0.519 (0.29)	0.290 (0.22)

Table 3: Coverage fraction, mean (std) over 64 seeds, under the C.2 configuration with the corrected log-space LPM. Best per column in bold: the uniform-random control wins *all three* variants. Every learned method falls below the no-learning control. LPM’s coverage is depressed by a bimodal collapse (29/64 seeds under `action_noise`; median 0.101); the collapse is policy instability, not TV-fixation (see text and Table 2).

Why the coverage ranking is not reproducible here (structural).

1. In a small maze (~ 1500 reachable cells) with no extrinsic reward, a random walk already covers most cells, so there is nothing for directed exploration to add; committing to any policy only narrows the walk. Intrinsic motivation targets *sparse-reward*, *hard-exploration* tasks (e.g. Montezuma’s Revenge), not pure coverage of a small room.
2. The maze notebooks contain only LPM and pixel-curiosity—*none* of Figure 3’s baselines (AMA, EDT, EME, Ensemble, IDF, RND). Figure 3 was produced by a fuller pipeline not in the released maze code; our RND/ICM are our own additions on the shared encoder.
3. Single-environment A2C at 5×10^4 steps is dominated by between-seed variance ($\text{std} \approx \text{mean}$)

and the curves have not plateaued at 50k; for LPM the variance is genuinely bimodal (collapse vs. explore) rather than merely noisy.

Spatial dynamics (RQ2). The 64-seed-averaged occupancy-density figure (Fig. 3, `action_noise`) makes the story legible once the shared log colour scale is used: brightness is the mean per-cell visit count. The *uniform* row spreads broadly and fairly evenly across both spawn rooms and up past the wall, at moderate per-cell counts; the trained policies instead *concentrate*—NONE and especially MSE show bright (10^2 – 10^3) dwell/spin cells where the agent loops in place, with sparser coverage elsewhere, while LPM/ICM/RND are patchier and dimmer (the collapsed LPM seeds contribute tight, bright single-cell loops that average into faint hotspots). This visualises directly why uniform covers more: training commits probability mass to a few revisited cells. The shared log scale is essential here—on the earlier per-panel linear scale these $1000\times$ dwell spikes saturated each panel and hid the surrounding path structure.

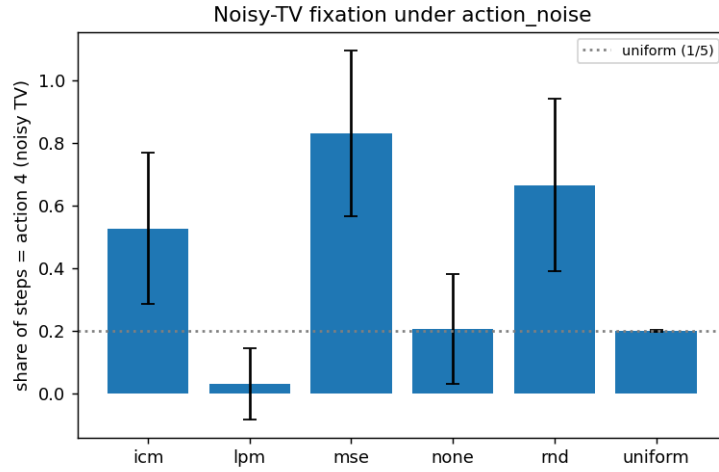


Figure 1: Share of steps spent on the noisy-TV action (action 4) under `action_noise`, mean $\pm$ std over 64 seeds. LPM $\approx$ 3.0%; MSE $\approx$ 83%; uniform = 1/5 (dotted line).

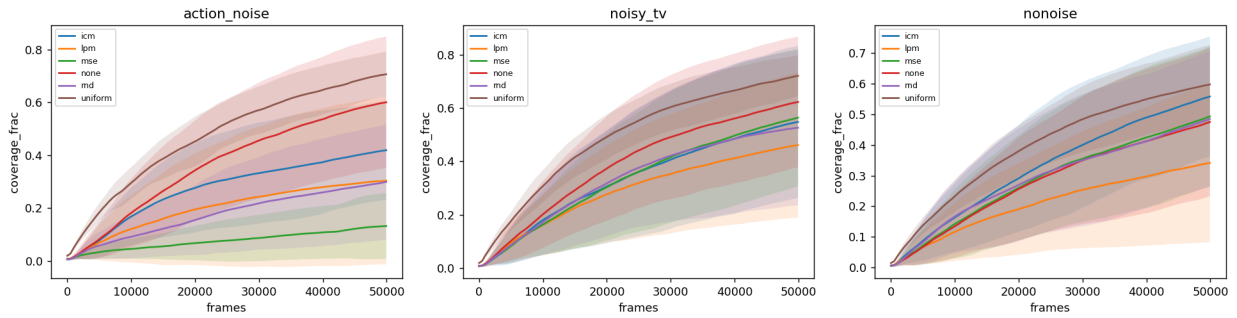


Figure 2: Reachable-cell coverage vs. frames, per variant (mean $\pm$ std over 64 seeds). The uniform-random control is highest in every panel; all trained policies fall below it.

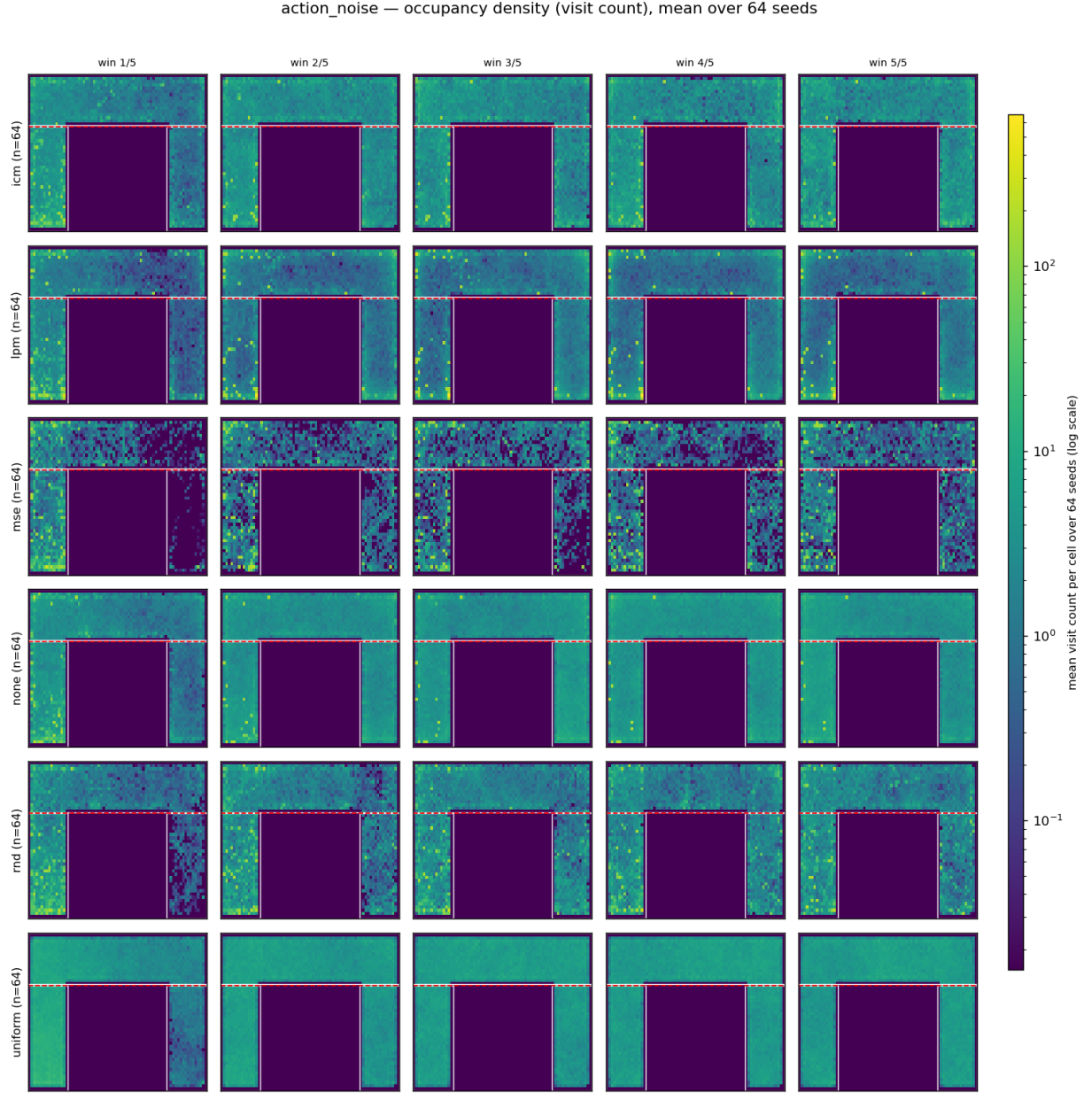


Figure 3: Occupancy-density evolution (`action_noise`), *averaged over all 64 seeds*. Rows: policy; columns: five equal training-time windows. Brightness is the mean per-cell visit count on a *shared log colour scale* (colorbar at right, $\sim 10^{-1}$ to $\sim 10^3$ visits); the red dashed line is the noise wall ($z = 8$), white rectangles are the reachable rooms. Uniform spreads broadly at moderate counts; MSE/NONE show bright concentrated dwell cells.

6 Infrastructure and reproducibility

The upstream maze experiment exists only as three Jupyter notebooks with an embedded A2C and no CLI. We extracted the A2C engine, the LPM prediction/uncertainty nets, and the pixel decoder *verbatim* into an importable, CLI-driven package (`LPM_exploration/Miniworld/experiments/`), aligned its hyperparameters to the paper’s Appendix C.2, **corrected the LPM intrinsic reward to the paper’s log-space Eq. (3)** (with the raw notebook form retained behind `reward_space="raw"`), added canonical ICM and RND on the shared CNN encoder, a `--random-policy` uniform control, and a per-step position log. The correction and the accompanying error-model learning-rate fix are covered by unit tests (`tests/test_models.py`). The environment geometry is reused from our own `miniworld_play/envs.py` (a faithful port of the upstream maze). Each run writes a per-update CSV (`coverage_frac`, `beyond_wall_frac`, `time_at_wall_frac`, plus model/policy losses) and a compact `.npz` of (step, x , z , action, sticky). A grid runner launches all runs in parallel (resumable); an analysis script aggregates the CSVs and position logs into the coverage table, the coverage curves, the TV-fixation figure, and the *seed-averaged* heatmap-evolution figures. All deviations are logged in `LPM_exploration/UPSTREAM.md`.

7 Limitations (stated up front)

1. **Single-environment variance.** We run the paper’s budget (5×10^4 steps) at 64 seeds, but each run is still a *single* environment, so coverage std stays at 0.08–0.32; the closer rankings among trained methods are illustrative, not significance-tested. The uniform-beats-all and TV-fixation results, however, are large and robust. LPM specifically is *bimodal* (explore vs. collapse), so its mean/median should be read together, not as a point estimate; a stability ablation (reward scaling / normalisation) is the natural follow-up.
2. **Coverage is the wrong yardstick here.** In a small, extrinsic-reward-free maze, coverage rewards randomness, so it cannot demonstrate the value of directed exploration—which is why uniform wins. A faithful test of LPM’s exploration benefit needs a sparse-reward, hard-exploration task (the paper’s Atari / Montezuma settings), where no-intrinsic agents get zero reward. What the maze *can* show—and does—is noise robustness, decoupled from coverage.
3. **Figure-3 baselines are absent from the released maze code.** The maze notebooks contain only LPM and pixel-curiosity; AMA/EDT/EME/Ensemble/IDF/RND are not present, so a full reproduction of the paper’s 7-method coverage ranking is not possible from this code. Our RND/ICM are our own ports.
4. **Coverage is a 2D projection.** Occupancy is binned on the floor plane; agent heading and the first-person view (what actually drives the pixel predictors) are not visualised here.